BRIDGE_C_35 order8

order8 REPORT (descriptive notes; verdicts as in RESULTS.md)

PREREG sha256 13b9df4f…f6d1 (unchanged). Script hashes: scripts/SHA256SUMS. No interior float span test is used as evidence. The only interior test is M3, a fixed-matrix residual with one free scalar, the same design as order5 Z. Deviations are in NOTES.md, and the verdicts below are the preregistered ones.

Category A (derived; sketches flagged)

- **Orthant angle.** The hyperbolic angle of the orthant between ĉ1 and ĉ′ is L = arccosh(7/(2√3.5)) = ½·log(6+√35). The two agree to 20 digits (mp), so the orthant is exactly half of the unit-group fundamental angle.
- **Map $\mathcal{H}$.** M1 uses $\mathcal{H}$ = D^{1/2} ∘ (y→∞ projection). The Laplace derivation of Θ_E = $\mathcal{H}$(Φ_orth) is a *sketch*.
- **Radial expansion of Φ_orth (Poisson).** t^{1/2}Φ_orth = (2π)^{−1/2}(2π/√35)·L·mean f + O(t^∞).
- **Character twist (S^2 algebra).** Under ρ_L ⊗ χ with χ(S) = ±i, the weight-1 law forces P-even. This is the P-altering candidate.

Category B (numerical)

map	test	statistic	verdict
M1: Θ_E = $\mathcal{H}$(Φ_orth), coefficient-wise	105 companions, all N ≤ 8, mp dps 25, y = 1e4/2e4 extrapolated	max |c_Φ − c_E| = 8.7e−10	**pass**
M2b/M2a: t^{1/2} Φ_orth is constant and equals the Poisson constant	x = 1 and ½, 2 companions, t = 2e−3 … 5e−4	constancy 2.5e−14; vs prediction 4.3e−14	**pass** (x = ½ is degenerate: mean f = 0)
M2c: radial A0, A1, A2 of Θ_E vs prediction (A0 from Φ via √(2/π); A1, A2 =	x ∈ {1, ½, ⅓}, 4 companions, preregistered 6-point grid, 5-term fit	A0 err ≤ 3.5e−8; A1 err ≤ 1.4e−5; A2 err 2.1e−5 … 9.5e−4	**FAIL** (threshold 1e−5)

map	test	statistic	verdict
edge/corner formula of the two shifted orthants)			
M3: twisted law $\Theta_E(-1/\tau) = c\cdot(\tau/\sqrt{35})\cdot M\cdot\Theta_E(\tau)$, with $c = \pm i$ required	660 τ, fixed M, free scalar c	c* = 0.001 + 0.080i, r = 0.54	**FAIL** (as predicted)
M3: same for the product-completed $\hat{E}$		c* = 0.002 − 0.701i, r = 0.375	**FAIL** (as predicted)
M3 control: $\hat{H}$		c* = 1.000, r = 7.5e−16	pass

Post-hoc check of M2c (descriptive, not a verdict; NOTES 1). I refit with 8 t-points and terms up to t^2. At x = 1 the max error is 3.7e−7, and at x = ⅓ it is 5.2e−6. This suggests the M2c failure comes from the under-resolved preregistered fit, not from the prediction. The recorded verdict remains FAIL.

Verdicts

- **M1 holds (Category B, Category A sketch).** The even double-orthant is exactly the image of the modular weight-0 Siegel object $\Phi_{orth} = \int_{\hat{c}1}^{\hat{c}'}\Theta_z\, d\mu$ under $\mathcal{H}$ (the half-derivative of the holomorphic projection). Φ_{orth} obeys the fixed weight-0 law $(1/\sqrt{35})\cdot M$ of the 35-component law vector. $\mathcal{H}$ is not modular-equivariant, so this is a mixed, non-modular link.
- **M2:** the Φ side passes. The Θ_E radial comparison **FAILS** at the preregistered resolution.
- **M3 (character twist / P-altering): FAILS.** Neither Θ_E nor $\hat{E}$ is modular under $\rho_L \otimes \chi$.
- **Full-series shadow (ξ-sense): NOT located.** By the preregistered rule, the even piece is identified as $\mathcal{H}(\Phi_{orth})$ (non-modular link), but the ξ-shadow of the full series is not located.

Appendix: order8 notes (disclosures; verdicts stand)

1. **M2c fit was under-resolved.** The preregistered 5-term fit on 6 points leaves out the t^{3/2} term, which biases A2 at about 1e−5 to 1e−3. m2posthoc.py is a new file, hashed before it ran, labelled descriptive. It refits with 8 points and terms up to t^2 for x = 1 and ⅓, giving max errors of 3.7e−7 and 5.2e−6. The M2c verdict stays FAIL.
2. **M2a/b at x = ½.** Here mean f = 0, so both the prediction and the measurement vanish (about 1e−15). That point is non-discriminating. x = 1 is the informative case.
3. **Interior M3 data.** M3 reuses order7's 660-τ arrays (read-only). The design is a fixed-matrix residual with one scalar, not a span test.
4. **Steering (3:42 PM CDT).** No weight −1 attack: odd weights are blocked by the parity obstruction.
5. **Reading.** The even piece has the form "D^{1/2} of the holomorphic projection of a weight-0 Siegel integral over the half unit-angle [ĉ1, ĉ′]". The natural next object is

the boundary of Φ_orth over the full unit arc [ĉ1, γĉ1], which is Φ_0 from order7: twice the orthant angle. That suggests Θ_E plus its γ-image (the complementary cone) equals $\mathcal{H}$(Φ_0), a full Hecke-type sum. Not tested here.

6. **Process.** nice, OMP_NUM_THREADS=1, ulimit -v 3000000. No push, no external sends. New files only.